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Editor-in-Chief
Applied Energy

Dear Editor,

We submit the manuscript “From single-event headroom to repeatable services: AC-audited recovery envelopes for distributed batteries in active distribution feeders” for consideration as a Full Length Research Article in Applied Energy.

The manuscript addresses a practical gap in distribution-system flexibility: a battery may pass a one-event power test yet fail the next call because recharge is constrained by feeder voltage and branch currents. We formulate a deadline-aware service-recovery contract, compose it over repeated calls, and audit candidate contracts with nonlinear OpenDSS power flow on held-out load/PV trajectories. The paper reports deliverable power, recovery success, sequence contraction and the network recovery penalty, with state-information sensitivity as an operating ablation.

The work is original, is not under consideration elsewhere, and has been approved by both authors. The authors declare no competing interests and no specific grant funding. An AI-use declaration is included in the manuscript. Code will be released under the MIT license at the public repository URL added before submission.

Sincerely,
Siqiang Zhao
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